Learning nuisances with kernels

1 Goal

The baseline kernel NPLM learns a flexible log-density ratio between data and a central reference sample. With nuisance parameters, the null hypothesis is composite. We write the nuisance-deformed reference density as

$$n(x \mid R_\nu) = r(x;\nu)\, n(x \mid R_0), \tag{1}$$

where R_0 is the central reference. The full alternative model can then be written as

$$h(x;\alpha,\nu) = f_K(x;\alpha) + \log r(x;\nu), \tag{2}$$

where f_K is the residual kernel NPLM component and $\log r(x;\nu)$ is the systematic deformation.

The first implementation step is to model $\log r(x;\nu)$ with kernel methods without modifying the existing baseline NPLM implementation.

2 Option A: Finite-Difference Falkon Morphing

Assume that nuisance-varied Monte Carlo samples are available at $\nu = +\epsilon$ and $\nu = -\epsilon$, together with a central sample at $\nu = 0$. Train two LogisticFalkon density-ratio estimators:

$$g_+(x) \simeq \log \frac{p(x \mid +\epsilon)}{p(x \mid 0)}, \tag{3}$$

$$g_-(x) \simeq \log \frac{p(x \mid -\epsilon)}{p(x \mid 0)}. \tag{4}$$

The local nuisance response is then estimated by the centered finite difference

$$\delta(x) = \frac{g_+(x) - g_-(x)}{2\epsilon}. \tag{5}$$

For one nuisance parameter, the morphing model is

$$\log r(x;\nu) \simeq \nu\, \delta(x). \tag{6}$$

For several independent nuisance directions, the first-order extension is

$$\log r(x;\nu) \simeq \sum_a \nu_a\, \delta_a(x). \tag{7}$$

This is the closest kernel analogue of the neural-network nuisance setup: the nuisance log-ratio is a separate component, while the final NPLM fit learns only the residual discrepancy $f_K(x)$.

2.1 Classifier Weighting

For central samples $x \sim p_0$ and varied samples $x \sim p_\nu$, a weighted binary logistic classifier has population minimizer

$$f^*(x) = \log \frac{N_\nu p_\nu(x)}{w_0 N_0 p_0(x)}. \tag{8}$$

To learn the normalized shape ratio p_ν/p_0, choose

$$w_0 = \frac{N_\nu}{N_0}. \tag{9}$$

If the nuisance also changes the expected total rate by a factor $\rho_\nu = N_\nu^{\text{expected}}/N_0^{\text{expected}}$, choose

$$w_0 = \frac{N_\nu}{N_0\,\rho_\nu}, \tag{10}$$

so that the score estimates

$$\log \rho_\nu \frac{p_\nu(x)}{p_0(x)}. \tag{11}$$

2.2 Caching

Falkon does not currently provide a simple native model serialization path. For the profiled NPLM statistic this is not a blocker, because the likelihood only needs repeated evaluations of $\log r(x;\nu)$ on fixed event arrays. After training the ratio estimators, cache

$$\delta_{\text{ref},i} = \delta(x_i^{\text{ref}}), \tag{12}$$

$$\delta_{\text{data},j} = \delta(x_j^{\text{data}}), \tag{13}$$

and save these arrays as a NumPy archive. Profiling then evaluates

$$\log r(x_i^{\text{ref}};\nu) = \nu\,\delta_{\text{ref},i}, \tag{14}$$

$$\log r(x_j^{\text{data}};\nu) = \nu\,\delta_{\text{data},j}. \tag{15}$$

This makes the nuisance layer cheap and independent of Falkon model persistence.

3 One-Dimensional Validation

The first validation toy uses a truncated exponential density on $[0, x_{\max}]$,

$$p(x \mid \lambda) = \frac{\lambda e^{-\lambda x}}{1 - e^{-\lambda x_{\max}}}. \tag{16}$$

The nuisance changes the rate as

$$\lambda(\nu) = \lambda_0 e^{a\nu}. \tag{17}$$

The analytic response at $\nu = 0$ is

$$\delta_{\text{true}}(x) = a - a\lambda_0 x - a\lambda_0 \frac{x_{\max} e^{-\lambda_0 x_{\max}}}{1 - e^{-\lambda_0 x_{\max}}}. \tag{18}$$

The implementation compares the Falkon finite-difference estimate $\delta(x)$ against this analytic curve.

4 Option B: Conditional Kernel Ratio

An alternative is to train a single conditional model on augmented inputs

$$z = (x, \nu) \tag{19}$$

so that

$$g(x, \nu) \simeq \log \frac{p(x \mid \nu)}{p(x \mid 0)}. \tag{20}$$

This can represent nonlinear nuisance dependence without a Taylor expansion. However, it requires careful kernel choices in the joint (x, ν) space, adequate coverage of the nuisance grid, and validation of interpolation in ν. For this reason the first implementation uses option A.

5 Current Implementation Boundary

The initial code lives in a separate `nplm_systematics` package. It does not modify the baseline `nplm.LogFalkonNPLM` class. The current scope is only the nuisance-ratio model

$$\log r(x; \nu), \tag{21}$$

not yet the full profiled statistic $t = \tau - \Delta$.